# Week 13: Route packing and bottlenecks

Adult guide built from the problem files · Unpiloted. Marker fit on the shortest arrows is unrehearsed.

This view repeats, by problem, the answers of the print guide, week-13-facilitator.pdf, whose source is editable/facilitator-src/build_guide.py; its page references are to that guide, apart from those that name a student page (see Page numbers). The overview, supplies, launch, proofs and hints are there. Change both together; check_guide_copy.py compares them.

## Problems and routes

| Problem | K–1 | 2–3 | 4–5 | route |
|---|---|---|---|---|
| Fit as many routes as you can | P1 | P1 | | first |
| Replace stuck reserved routes | | | P1 | first |
| Stop every route with as few markers as you can | P2 | | | first |
| The most routes, then stop all travel | P3 | | | first |
| The interior bottleneck | | P2 | P2 | first |
| Close one arrow and still fit two routes | P4 | | | return |
| Every pair of closing arrows on the bowtie | P5 | P3 | | K-1 return; 2-3 first |
| Add one arrow so that three routes fit | | P4 | | return |
| Replace stuck reserved routes | | P5 | | return |
| Splits with two cut arrows and an arrow back | | | P3 | first |
| A route that uses two closing arrows | | | P4 | return |
| A change-walk that adds a route | | | P5 | return |
| Every dot a change-walk can reach | | | P6 | return |
| No change-walk means largest | | | P7 | return |
| The closing game | P6 | P6 | | return |

**First route.** K–1 P1–P3; Grades 2–3 P1–P3; Grades 4–5 P1–P3. **Return visits.** K–1 P4–P6; Grades 2–3 P4–P6; Grades 4–5 P4–P7.

**Page numbers.** A key that names a student page counts the pages of the full band packet. The route packets keep the problem numbers but number their own pages; each route-packet page is this page of the full packet:

| Route packet | its pages = full-packet pages |
|---|---|
| K–1 first | pp. 1–3 = pp. 1–3 |
| K–1 return | p. 1 = p. 1; pp. 2–4 = pp. 4–6 |
| Grades 2–3 first | pp. 1–3 = pp. 1–3 |
| Grades 2–3 return | p. 1 = p. 1; pp. 2–5 = pp. 4–7 |
| Grades 4–5 first | pp. 1–4 = pp. 1–4 |
| Grades 4–5 return | p. 1 = p. 1; pp. 2–5 = pp. 5–8 |

## Key, one entry per problem

### Fit as many routes as you can

K–1 P1 · Grades 2–3 P1 · first route · wording differs by band

*Idea.* A stuck collection need not be largest: on the greedy board the route through the middle arrow blocks both others, and two routes fit.

**K–1:** Upper greedy board: maximum 2, uniquely the collection s-A-t and s-B-t. The route s-A-B-t alone is stuck but not largest. Lower bowtie: maximum 2, with exactly two collections: {s-A-C-D-t, s-B-C-E-t} or {s-A-C-E-t, s-B-C-D-t}.

Both may meet at C. Closing sA and sB blocks all travel and certifies 2.

**Grades 2–3:** Both boards have 2 routes and require 2 closures. Upper bowtie: take s-A-C-D-t and s-B-C-E-t; close {sA,sB}. Lower greedy: take s-A-t and s-B-t; close {sA,sB}. Page 4 explains the certificate; shared C does not consume capacity.

**K–1:** Hint: can each arrow out of Start serve a different route? If a chosen route blocks another, try changing the first.

**Grades 2–3:** Hint: distinguish a dot that is shared from an arrow that is shared.

## Replace stuck reserved routes

Grades 4–5 P1 · first route

*Idea.* Reserved routes that leave no room can be replaced by a larger collection; on the double-trap board two reservations must be undone at once.

First board: replace the reserved s-A-B-t by s-A-t and s-B-t. The change-walk s-B-A-t uses unused sB and At and cancels AB by traversing it backward. Two Start arrows certify the optimum of 2.

Second board is the revised coupled network, with the reserved route s-A-B-C-D-E-t shown thick above. Its maximum is 2, not 3. Two final collections are {s-A-C-D-t, s-B-C-E-t} and {s-A-C-E-t, s-B-C-D-t}. No third can leave Start.

First ask whether both arrows leaving Start could be used. Then ask which reservation prevents the other approach from continuing. If that repair still leaves only one route, inspect the other half. Do not call a backward change-step a legal final route arrow.

## Stop every route with as few markers as you can

K–1 P2 · first route

*Idea.* Every route must lose an arrow: the diamond needs one marker on each branch, and the funnel needs one on the arrow every route shares.

Upper diamond: minimum 2. All four minimum closures take one arrow from {sA,At} and one from {sB,Bt}: {sA,sB}, {sA,Bt}, {At,sB}, {At,Bt}. Each branch must be broken. Lower funnel: minimum 1; either CD or Dt stops both routes. There are no other one-arrow solutions because closing an earlier edge leaves the other branch.

Hint: trace both routes and look for an arrow they must both use.

## The most routes, then stop all travel

K–1 P3 · first route

*Idea.* The most routes equals the fewest closing markers: two routes and two markers after the meeting dot, three routes and three markers at Start.

Upper board below: 2 routes, for example s-A-D-E-t and s-B-D-F-t; close DE and DF. Lower three-branch board: 3 routes, s-A-t, s-B-t and s-C-t; close sA, sB and sC. Every route uses one selected outgoing Start arrow. The page asks for as few markers as possible; ask children to compare that number with the number of routes.

Hint: a shared dot may have several incoming arrows but fewer outgoing arrows.

## The interior bottleneck

Grades 2–3 P2 · Grades 4–5 P2 · first route · wording differs by band

*Idea.* Three arrows leave Start and three enter Finish, but every route must use one of the two arrows leaving D, so two routes and two closures certify each other.

**Grades 2–3:** Maximum and minimum are both 2. Use s-A-D-E-G-H-t and s-C-D-F-G-J-t; close DE and DF (or EG and FG). Start has three outgoing arrows and Finish three incoming arrows, but every route must pass through one of the two arrows leaving D. Hence no collection of three fits. The board has 18 individual routes and 36 unordered maximum collections, but listing them is unnecessary. See the reference diagram on page 7.

**Grades 4–5:** Use the interior bottleneck solution on page 7: two routes and two closures, DE and DF. Every s-to-t route must leave the start-side set {s,A,B,C,D}, and these are its only outgoing arrows. The certificate argument on page 4 supplies both required inequalities.

**Grades 2–3:** Hint: look in the middle, not only at the ends.

**Grades 4–5:** Hint: explain why all routes cross the proposed barrier before claiming the count is optimal.

## Close one arrow and still fit two routes

K–1 P4 · return route

*Idea.* Closing an arrow before the meeting dot leaves two routes; closing one of only two exits, or a branch of the greedy board other than its middle arrow, does not.

On the upper board, exactly six single closures preserve two routes: sA, sB, sC, AD, BD, CD. Each removes one of three approaches to D, leaving two intact. Any of DE, DF, Et or Ft removes one of only two exits and leaves at most one route. On the lower greedy board, only AB may be removed. Any other closure breaks one of the two disjoint branch routes. Reopen the arrow before each new trial.

Hint: sort arrows into “before the meeting dot” and “after it.”

## Every pair of closing arrows on the bowtie

K–1 P5 · Grades 2–3 P3 · K-1 return; 2-3 first route · wording differs by band

*Idea.* Exactly eight pairs stop all travel: one half of the bowtie must be blocked completely, with one closure on each of its two branches.

All eight pairs of closing arrows on the bowtie are below. Here brackets mean “choose one from each listed pair of alternatives.” Before C, choose one from {sA,AC} and one from {sB,BC}. After C, choose one from {CD,Dt} and one from {CE,Et}.

| Before C | After C |
|---|---|
| {sA,sB} | {CD,CE} |
| {sA,BC} | {CD,Et} |
| {AC,sB} | {Dt,CE} |
| {AC,BC} | {Dt,Et} |

Completeness: if a path from s to C and a path from C to t both survive, concatenate them to get a whole route. So at least one half must be entirely blocked. Each half has two independent branches; blocking it with exactly two arrows requires one closure on each branch. A mixed pair with one arrow on each half cannot block either half.

**K–1:** Exactly eight pairs; use the list and completeness proof on page 6. K-1 page 5 has a large working board and ten small recording copies, so the copies do not give the count away. Mark a different successful pair on each; a marker rearrangement alone does not preserve the record.

Hint ladder: cover the right half temporarily; ask what it takes to stop reaching C; then repeat on the other half.

**K–1:** Hint: check that two marked arrows belong to the same pair before comparing records.

## Add one arrow so that three routes fit

Grades 2–3 P4 · return route

*Idea.* A new inside arrow helps only if it jumps past both two-arrow bottlenecks; on the board with two Start arrows no inside arrow can help.

Upper board: any new arrow from A, B, C or D to G, H, I or J works (16 choices); it must jump past both two-arrow bottlenecks. With D to G, three routes are s-A-D-E-G-H-t, s-B-D-G-I-t, and s-C-D-F-G-J-t. They meet at D and G but share no arrows. Lower board: impossible. Start has only two outgoing arrows, and the allowed new arrow cannot touch Start. Every collection is therefore bounded by 2.

Hint: before searching, ask whether there is a bottleneck the allowed move cannot change.

## Replace stuck reserved routes

Grades 2–3 P5 · return route

*Idea.* Reserved routes that leave no room can be replaced by a larger collection; on the double-trap board two reservations must be undone at once.

First board: replace the reserved s-A-B-t by s-A-t and s-B-t. The change-walk s-B-A-t uses unused sB and At and cancels AB by traversing it backward. Two Start arrows certify the optimum of 2.

Second board is the revised coupled network, with the reserved route s-A-B-C-D-E-t shown thick above. Its maximum is 2, not 3. Two final collections are {s-A-C-D-t, s-B-C-E-t} and {s-A-C-E-t, s-B-C-D-t}. No third can leave Start.

First ask whether both arrows leaving Start could be used. Then ask which reservation prevents the other approach from continuing. If that repair still leaves only one route, inspect the other half. Do not call a backward change-step a legal final route arrow.

## Splits with two cut arrows and an arrow back

Grades 4–5 P3 · first route

*Idea.* Every route leaves the Start side at a first exit through an outgoing cut arrow, so closing the outgoing cut arrows stops travel; arrows pointing back do not count.

Any two of these three start-side sets work. All other dots are on the Finish side. They are the complete list satisfying both requested conditions.

| Start-side set | Outgoing cut arrows | Arrows entering Start side |
|---|---|---|
| {s,A} | sB, AC | BA |
| {s,A,B} | AC, BD | CB |
| {s,A,B,D} | AC, Dt | CD, CB |

Every directed route begins on the Start side and ends outside it, so at its first exit it must use an outgoing cut arrow. Closing all outgoing cut arrows therefore blocks travel. Incoming arrows do not allow an exit and are not counted. A route can later reenter the side, but that cannot avoid its first exit.

Hint: color the dots by side, then follow each arrowhead across the color boundary.

## A route that uses two closing arrows

Grades 4–5 P4 · return route

*Idea.* A route may cross a smallest cut more than once, so every route uses at least one cut arrow, not exactly one, and such a route cannot be in a largest collection.

Close AC and BD. This is a minimum two-arrow set: s-A-C-t and s-B-D-t are disjoint, so one closure is insufficient. The route s-A-C-B-D-t uses both AC and BD. It cannot belong to a largest collection: every other route must use AC or BD, both already reserved. The largest size is 2, but any collection containing this long route has size 1. This example refutes the claim that every route uses exactly one outgoing cut arrow; “at least one” is the correct bound. There are

five smallest sets: {sA,sB}, {sB,AC}, {AC,BD}, {AC,Dt}, {Ct,Dt}. Only the middle three are used twice by one route (s-B-A-C-t, s-A-C-B-D-t, s-A-C-D-t). If a child chose the two Start or the two Finish arrows, ask for a different smallest set. The answer to the question is always no.

Hint: reserve the long route and test which cut arrows remain available.

## A change-walk that adds a route

Grades 4–5 P5 · return route

*Idea.* A change-walk may follow unused arrows forward and reserved arrows backward; toggling its arrows turns two stuck routes into three.

The reserved routes are s-B-D-t and s-C-E-t. Use the change-walk s-A-D-B-E-C-F-t. The steps D to B and E to C cancel the original BD and CE. Other steps reserve sA, AD, BE, CF, Ft. The resulting three forward routes are s-A-D-t, s-B-E-t and s-C-F-t. All arrows are present in the printed board. The three arrows leaving Start bound every collection by 3, so this improvement is optimal. The problem also asks for the two Problem 1 boards: their change-walks are s-B-A-t and s-B-A-C-E-D-t (page 7).

The change-walk is forced here: only sA is initially available from Start; after AD, the used Dt is unavailable forward, so go backward along BD; continue analogously through E. This is a different interaction from the revised bowtie case.

Hint: redraw only the currently available forward steps and allowed cancellation steps, keeping their meanings distinct.

## Every dot a change-walk can reach

Grades 4–5 P6 · return route

*Idea.* With two routes reserved on the bottleneck board, change-walk steps reach exactly s, A, B, C, D, and the arrows leaving that set are a closing set as large as the collection.

The printed reservations on the bottleneck board are s-A-D-E-G-H-t and s-C-D-F-G-J-t. Exactly s,A,B,C,D are reachable by change-walk steps. A search witness: s-B-D reaches B and D; backward steps D-A and D-C then reach A and C. No change-step leaves that set because DE and DF are reserved and point outward; their backward directions start outside the set.

Thus {DE,DF} is a two-arrow closure matching the two reserved routes. No other dot is reachable: E,F,G,H,I,J,t all lie beyond those saturated arrows, and there is no unused outgoing crossing.

Hint: mark reached dots, examine every allowable step leaving one, and continue until no new dot can be marked. Do not stop after one failed branch.

## No change-walk means largest

Grades 4–5 P7 · return route

*Idea.* If no change-walk reaches Finish, the arrows leaving the reachable set are all reserved and number exactly k, so closing them proves the k routes are largest (max-flow min-cut).

Answer: yes. Suppose a collection of k routes has no change-walk to Finish. Mark S, all vertices reachable from Start by unused forward arrows and used backward arrows. Start is in S and Finish is outside. Every original arrow leaving S is reserved, or its unused forward direction would reach another dot. Every original arrow entering S is unreserved, or its backward direction would reach a dot outside S.

At every internal dot, the number of reserved arrows coming in equals the number going out. Add this balance over all dots of S. Internal arrows cancel in pairs. The only surplus is k at Start, since Finish is outside. Hence k equals the number of reserved outward arrows minus the number of reserved inward arrows. The latter is zero; all outward arrows are reserved. There are therefore exactly k outward arrows, and closing them blocks every route. Page 4's certificate proves the current collection is largest.

## The closing game

K–1 P6 · Grades 2–3 P6 · return route · wording differs by band

*Idea.* On the diamond the second player wins by breaking the other branch; on the greedy board the first player wins only by closing the middle arrow first.

Upper diamond: choose second. Any first move breaks one branch; break the other to win immediately. Lower greedy board: choose first and close AB. This leaves the diamond for the opponent, so break the remaining branch after their move. This is the only winning opening: any other first closure leaves all surviving routes sharing one arrow, which the opponent closes at once.

Hint: play each opening with roles reversed; ask who can make the last necessary closure, not who closes the most arrows.